CHAPTER 8

THE FIRST NEW PLANET

I implore you, you do not hope to be able to give the reasons for the number of planets, do you?

This worry has been resolved …

—JOHANNES KEPLER,

EPITOME OF COPERNICAN ASTRONOMY,

BOOK 4 (1621)

Before we invented civilization, our ancestors lived mainly in the open, out under the sky. Before we devised artificial lights and atmospheric pollution and modern forms of nocturnal entertainment, we watched the stars. There were practical calendrical reasons, of course, but there was more to it than that. Even today, the most jaded city dweller can be unexpectedly moved upon encountering a clear night sky studded with thousands of twinkling stars. When it happens to me, after all these years, it still takes my breath away.

In every culture, the sky and the religious impulse are intertwined. I lie back in an open field and the sky surrounds me. I'm overpowered by its scale. It's so vast and so far away that my own insignificance becomes palpable. But I don't feel rejected by the sky. I'm a part of it—tiny, to be sure, but everything is tiny compared to that overwhelming immensity. And when I concentrate on the stars, the planets, and their motions, I have an irresistible